

Hourglass-Shaped Nanocages with Concaved Structures Based on Selective Self-Complementary Coordination Ligands and Tunable Hierarchical Self-Assembly

Yaping Xu, Haixin Zhang, Haoyue Su, Jianjun Ma, Hao Yu, Kehuan Li, Junjuan Shi, Xin-Qi Hao, Kun Wang,* Bo Song,* and Ming Wang*

Three-dimensional (3D) structures constructed via coordination-driven self-assemblies have recently garnered increasing attention due to the challenges in structural design and potential applications. In particular, developing new strategy for the convenient and precise self-assemblies of 3D supramolecular structures is of utmost interest. Introducing the concept of self-coordination ligands, herein the design and synthesis of two meta-modified terpyridyl ligands with selective self-complementary coordination moiety are reported and their capability to assemble into two hourglass-shaped nanocages SA and SB is demonstrated. Within these 3D structures, the meta-modified terpyridyl unit preferably coordinates with itself to serve as concave part. By changing the arm length of the ligands, hexamer (SA) and tetramer (SB) are obtained respectively. In-depth studies on the assembly mechanism of SA and SB indicate that the dimers could be formed first via self-complementary coordination and play crucial roles in controlling the final structures. Moreover, both SA and SB can go through hierarchical self-assemblies in solution as well as on solid–liquid interface, which are characterized by transmission electron microscope (TEM) and scanning tunneling microscopy (STM). It is further demonstrated that various higher-order assembly structures can be achieved by tuning the environmental conditions.

emergence of a large number of two-dimensional (2D)^[1–12] (such as polygons, concentric polygons, and tessellated architectures) and three-dimensional (3D) supramolecular architectures^[13–21] (such as prisms, polyhedron, truncated polyhedron, and a variety of other cages). In particular, 3D structures have received extensive attention and led to applications in catalysis,^[22,23] sensing,^[24,25] and drug delivery.^[26,27] However, developing new ligands which further contribute to the construction of 3D architectures with novel strategies remains a challenge.

Among many reported organic ligands, 2,2':6',2''-terpyridine (tpy) has attracted numerous attentions^[28–34] due to its rich coordination sites and stronger binding energy as compared to pyridyl and bipyridyl ligands. The pseudo-octahedral coordination <tpy-M(II)-tpy> with a large variety of transition metals has not only provided candidates with abundant connectivity sites, but it also enabled the formation a rigid linear linkage between the two terpyridyl units. As such, a series of 2D

metallo-supramolecular architectures has been constructed^[35–42] in the past three decades. However, due to the 180° angle of motif <tpy-M(II)-tpy>, the construction of 3D structures using terpyridyl ligand with substituent groups on the 4' position is limited. To date, the construction of terpyridyl 3D structures mainly relies on the inherent angle of organic backbones, such

1. Introduction

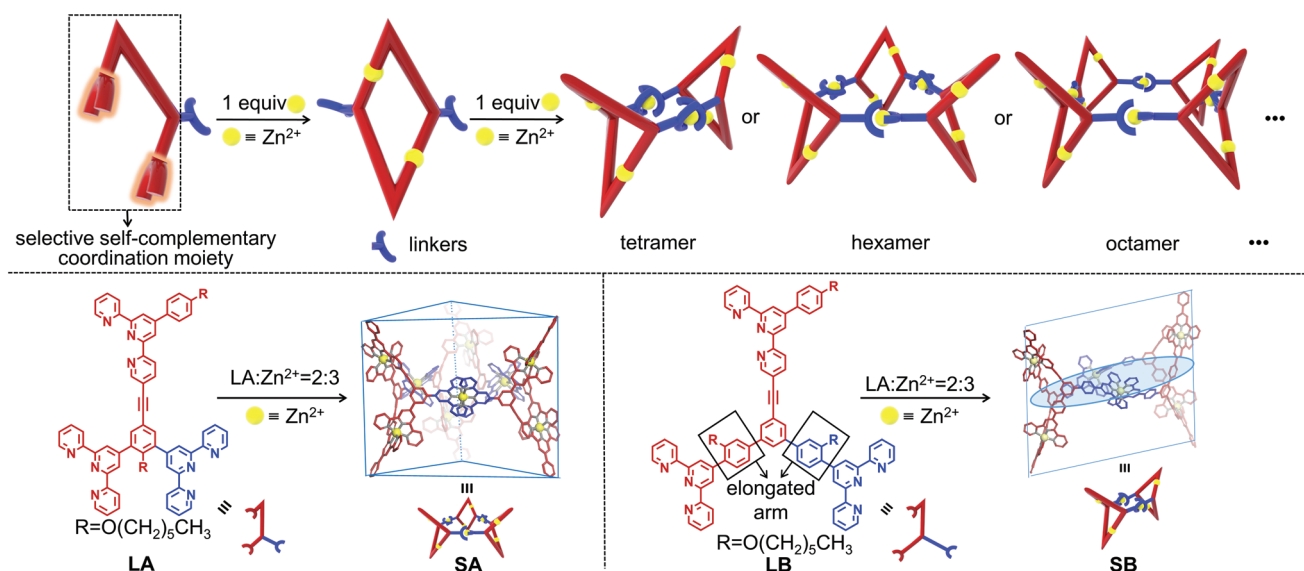
Coordination-driven self-assembly has been a powerful approach to construct sophisticated and elegant structures due to its excellent directionality and controllability of metal–ligand coordinative bonds. The past few decades have witnessed the

Y. Xu, H. Su, J. Ma, H. Yu, K. Li, J. Shi, M. Wang
State Key Laboratory of Supramolecular Structure and Materials
College of Chemistry
Jilin University
Changchun, Jilin 130012, China
E-mail: mingwang358@jlu.edu.cn
H. Zhang, K. Wang
Department of Physics and Astronomy
Mississippi State University
Mississippi State, MS 39762, USA

X.-Q. Hao
College of Chemistry and Green Catalysis Center
Zhengzhou University
Zhengzhou, Henan 450001, China
K. Wang
Department of Chemistry
Mississippi State University
Mississippi State, MS 39762, USA
E-mail: kw2504@msstate.edu
B. Song
Department of Chemistry
Northwestern University
Evanston, IL 60208, USA
E-mail: bosong@northwestern.edu

 The ORCID identification number(s) for the author(s) of this article can be found under <https://doi.org/10.1002/sml.202300009>.

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Scheme 1. Synthetic strategy of SA and SB.

as controlling ligand geometries by introducing meta- or ortho-substituted phenyl spacers^[43–47]; as well as using special vertices (adamantane,^[48] triptycene,^[49] 9,10-ethanoanthracene,^[50] corannulene,^[51] crown ether,^[52,53] etc.). Due to the scarcity of ligand with appropriate angles, designing new strategies for the construction of 3D metallo-supramolecules based on <tpy-M(II)> connections are highly required.

One way to broaden the scope of terpyridine-based self-assembly is to utilize the subunits assembled by organic ligands with metal ions as promising building blocks for further self-assembly with the uncoordinated terpyridyl units. It is expected that harnessing the angular freedom provided by self-assembled subunits, which overcomes the limitation imposed by the inherent angles of organic ligands, will greatly expand the diversity and tunability of terpyridyl 3D structures. The key to this strategy is the design of suitable ligand that can accurately coordinate with metal ions to obtain the expected subunit with convergent angle. To address this problem, we introduce the concept of self-coordination ligands. Selective self-complementary coordination terpyridyl ligands have great potential for precise and controllable self-assemblies of supramolecular structures. We predict that meta-modified terpyridyl ligands can self-coordinate with metal ions to form dimer subunits. Using the dimer subunits as fundamental building blocks, various structures with specific shapes and sizes could be synthesized in principle, such as tetramer, hexamer, octamer, decamer, etc. (**Scheme 1**). To the best of our knowledge, such ligand design strategy has rarely been reported.

Herein, we synthesized two dissymmetrical tritopic terpyridyl ligands LA and LB, which were self-assembled with Zn(II) in a 2:3 ratio, forming the hourglass-shaped nanocages SA and SB, respectively. In-depth mechanistic study revealed that during the self-assembly process both LA and LB went through self-complementary coordination process, producing dimers DA and DB, which were subsequently self-assembled with Zn(II) to form the desired hexamer and tetramer. Moreover, the hierarchical self-assembly behavior of these supramolecular

structures was also systematically studied. We find that the hydrophobic interactions together with π - π interactions drive the hierarchical self-assemblies into different morphologies: SA and SB could self-assemble into hollow rings and nanowires respectively in a mixed solvent of DMF and THF. After reducing the polarity of the solvent, the hierarchical assembly morphology of SA and SB showed nanosheets and rod-like structures. When SA and SB were drop casted on surface, nanowires could be clearly observed using scanning tunneling microscopy (STM).

2. Results and Discussion

2.1. Synthesis, Self-Assembly, and Characterization of SA

To construct 3D structure with preferential self-complementary coordination dimer as edge, we first designed and synthesized a terpyridyl compound **9** with a bromine attached to the 5'' position using Kröhnke reaction (**Scheme S1**, Supporting Information).^[54] Afterward, the dissymmetrical tritopic ligand LA was achieved in a decent yield of 68% by connecting a 120° bis-terpyridyl motif with compound **9** using Sonogashira coupling reaction. LA was fully characterized by nuclear magnetic resonance (NMR) spectroscopy and matrix-assisted laser desorption-ionization time of flight mass spectrometry (MALDI-TOF-MS) analysis (**Figure S1**, Supporting Information). The characteristic proton resonances were assigned based on 2D NMR spectroscopy (COSY and NOESY) (**Figures S4–S7**, Supporting Information).

After the successful synthesis of ligand LA, we started our journey toward the self-assembly process. LA was initially mixed with $\text{Zn}(\text{NO}_3)_2 \cdot 6\text{H}_2\text{O}$ in a stoichiometric ratio (2:3) in a mixed solution of chloroform/methanol (v/v = 1:3) at 50 °C for overnight. Then, an excess amount of NH_4PF_6 was added, and a yellow precipitate was formed which was isolated via centrifugation. The obtained product was first characterized

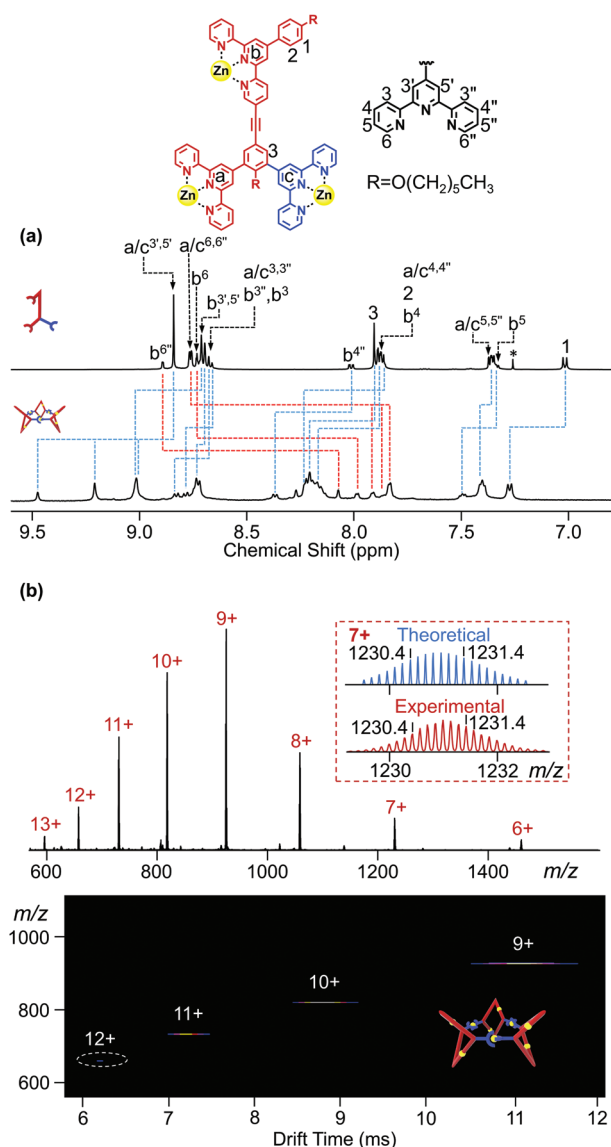


Figure 1. a) ¹H NMR spectra (500 MHz, 300 K) of LA in CDCl₃ and SA in CD₃CN/CD₃OD (4:1, v/v). b) ESI-MS and TWIM-MS plots (*m/z* vs drift time) of SA.

via ¹H NMR. As shown in **Figure 1a**, the proton resonances of complex SA were slightly broadened due to the slow tumbling rate after coordination, indicating the formation of a large complex.^[55] Moreover, the significant upfield and downfield shifts compared to LA also indicates the successful coordination with Zn(II). Protons on the 6, 6'' position of all tpy subunits showed upfield shifts with $\Delta\delta \approx 0.82$ (b⁶), 0.75 (b^{6'}), 0.80 (a^{6,6''}), and 0.87 (c^{6,6''}), respectively, owing to the electron shielding effect after coordination. Meanwhile, protons on all other positions of tpy units showed downfield shifts, owing to the electron deficiency of these protons. It is noteworthy that the protons in tpy-H^{a/c3,3'} position split into three sets of signals with an integration ratio of 1:1:2 in SA. This indicates that the two terpyridyl groups on the 120° bis-terpyridyl subunit served as two different roles during the self-assembly, leading to a dissymmetric chemical

environment. Similarly, the proton of tpy-H^{a/c6,6''} also split into three sets of doublet signals with an integration ratio of 1:1:2 in SA, which were assigned to tpy-H^{a6}, tpy-H^{a6''}, and tpy-H^{c6,6''}, respectively. This is possibly due to the steric hindrance of the bulky <tpy-Zn(II)-tpy> subunit and self-complementary coordination which restricted free rotation of tpy-a.

Furthermore, the chemical composition of SA was confirmed by electrospray ionization mass spectrometry (ESI-MS). As shown in **Figure 1b**, a series of peaks with continuous charge states from 6+ to 13+ were observed due to the losing of different numbers of PF₆⁻ counterions. The isotope pattern of each charge state matched well with theoretical simulation (**Figure S23**, Supporting Information). Deconvolution of the mass-to-charge ratios provided a molecular weight of 9613.6 Da, indicating the successful formation of hexamer with chemical composition of [LA₆Zn₆]¹⁸⁺PF₆⁻. Moreover, travelling wave ion mobility mass spectrometry (TWIM-MS) showed a set of signals in which each *m/z* value correspond to only one drift time distribution, indicating a single discrete complex without other isomers or structural co-conformations. Meanwhile, the diffusion-ordered NMR spectroscopy (DOSY) of SA (**Figure 3a**) showed one single band for all the resonances with a diffusion coefficient of $\log D = -9.52$, revealing the formation of a single discrete structure. The experimental hydrodynamic radius of SA was calculated as ≈ 2.0 nm via the Stokes–Einstein equation.

2.2. Synthesis, Self-Assembly, and Characterization of SB

After the successful assembly of SA, we continued our journey toward the synthesis of supramolecule SB. While the self-assembly using LA gave SA as an hourglass-shaped nanocage with inner cavity of 10817 Å³, we speculate whether the ligand with elongated arms would give a supramolecular self-assembly with a larger cavity. Consequently, ligand LB with extended phenylene skeleton was designed and synthesized through Suzuki coupling reactions and was obtained in a good yield (70%). Following the same synthetic procedure as SA, SB was synthesized after mixing LB with Zn(II) in stoichiometric ratio. The ¹H NMR of SB showed characteristic upfield shifts of all tpy-H^{6,6''} due to the electron shielding effect. Similar to SA, the proton of tpy-H^{a/c3,3''} in SB were obviously split into three sets of doublet resonances with integration ratio of 1:1:2, indicating that the two terpyridines on the 120° arm played two different roles during self-assembly. Also, other protons of tpy-a showed much broader signals compared with tpy-c. This phenomenon may be ascribed to the congested space caused by not only the self-complementary coordination, but also the existence of the alkyl chains on the adjacent aromatic rings of tpy-a, which restricts the free rotation of the tpy-a group and slows down the tumbling motion of tpy-a. The specific proton attribution was confirmed by COSY and NOESY (**Figures S19–S22**, Supporting Information).

ESI-MS was further utilized to characterize SB. As shown in **Figure 2b**, the mass spectrum of SB displayed a series of clean signals with charge states ranging from the 5+ to 11+ on account of losing different PF₆⁻ counterions. To our surprise, the deconvolution of the signals gave a molecular weight of 7417.7 Da, which corresponded to a tetrameric chemical

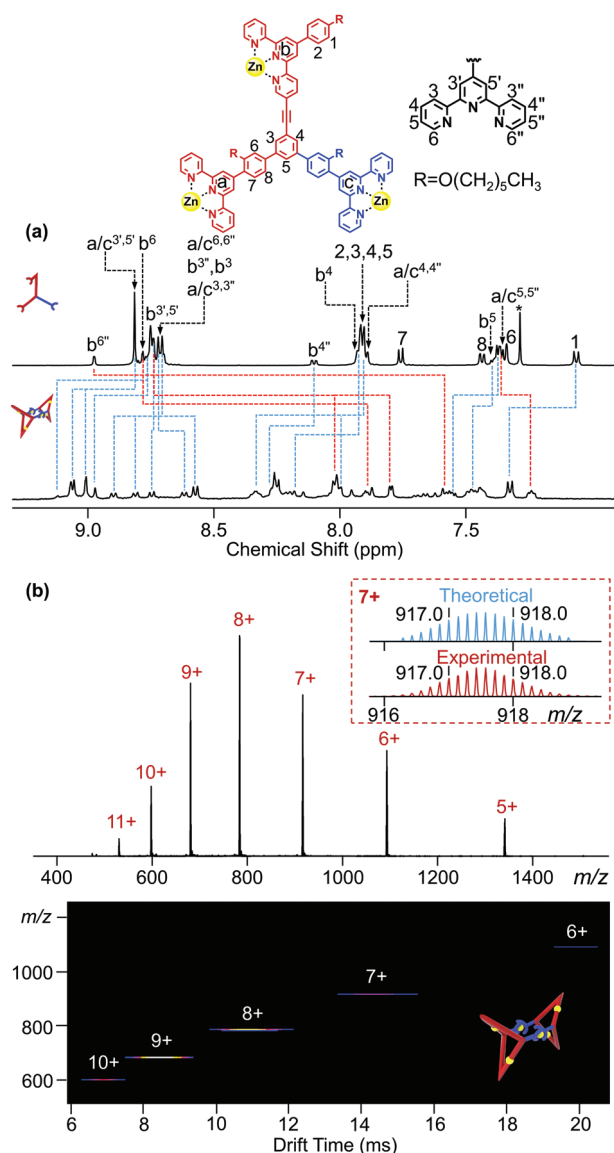


Figure 2. a) ^1H NMR spectra (500 MHz, 300 K) of LB in CDCl_3 and SB in CD_3CN . b) ESI-MS and TWIM-MS plots (m/z vs drift time) of SB.

composition of $[\text{LB}_4\text{Zn}_6] \cdot 12\text{PF}_6^-$. Meanwhile, the isotopic pattern of each peak agreed well with theoretical simulations (Figure S24, Supporting Information). Moreover, TWIM-MS showed only one set of signals, implying the formation a single discrete assembly. DOSY NMR spectrum (Figure 3b) also showed all resonances at a single band with diffusion of $\log D = -9.60$, which corresponds to a larger hydrodynamic radius (2.41 nm) as compared to SA.

2.3. Study of the Self-Assembly Mechanism

The structural difference between SA and SB intrigued us to perform an in-depth investigation into the self-assembly mechanism. According to our previous speculation, both ligands LA and LB are self-complementary and are expected

to coordinate with Zn(II) to form a dimer, and then self-assemble into the corresponding supramolecules. To verify the proposed route, we first mixed LA and Zn(II) in equal molar ratio, and the mixed sample was then heated at 50 °C overnight. After counterion exchange into PF_6^- followed by centrifugation, a yellow powder, namely DA, was obtained. DA was characterized via ESI-MS and TWIM-MS. The mass spectrum (Figure 4c, Figure S27, Supporting Information) of DA showed three positively charged signals (losing two, three, and four PF_6^- counterions, respectively), which is consistent with the chemical composition of $[\text{LA}_2\text{Zn}_2] \cdot 4\text{PF}_6^-$ with a molecular weight of 2850.70 Da. TWIM-MS only showed one set of signals with narrow drift time distributions (Figure 4d), which excluded the formation of other constitutional isomers. This result suggests that the formation of DA is favored during the self-assembly process. Similar to LA, LB could also self-assemble with Zn(II) in 1:1 ratio to form dimer DB with a molecular weight of 3355.01 Da, which was also verified by ESI-MS and TWIM-MS (Figure 4g,h and Figure S28, Supporting Information). Consequently, we speculated that both LA and LB were firstly self-assembled as a dimer DA and DB during self-assemblies, followed by coordination among the free linkers which generated the self-assembly of hexamer and tetramer, respectively. Moreover, molecular dynamics studies yielded DA and DB as two saddle-shaped structures. As illustrated in Figure 4b, the two free terpyridines units on DA opens parallelly with a dihedral angle of 56.4°. The distance between the two free terpyridines were measured to be 1.3 nm. Such close distance might have resulted in steric hindrance and facilitated the self-assembly into hexamer instead of tetramer. In contrast, the two free terpyridine units on DB are staggered and the dihedral angle was measured to be 64.3°, while the distance between the two free terpyridines was 2.0 nm (Figure 4f). Also taking into consideration that the elongated arms herein provide more flexibility, the free terpyridines in DB tend to coordinate in a complementary manner to produce SB as a tetramer.

2.4. Investigation of the Hierarchical Self-Assembly Behavior

To gain an in-depth understanding of the structures of SA and SB, we investigated their hierarchical self-assembly behavior using several microscopy methods. We initially realized that SA and SB behaved differently in acetonitrile solution upon changing concentrations. While the solution of SA remained mobile even at a high concentration of 30 mg mL^{-1} , the solution of SB switched into gel immediately at a concentration of 24 mg mL^{-1} (Figures S29 and S30, Supporting Information). We speculate that such difference might result from the difference in the structures, and that SA and SB could also behave in a different manner during the hierarchical self-assemblies into higher-ordered structures.

We first studied the hierarchical behavior in solution. By slowly evaporating tetrahydrofuran (THF) into a *N,N*-dimethylformamide (DMF) solution of SA over a period of 2 days, a suspension was obtained which was drop-casted on the carbon-coated Cu grid for transmittance electron microscopy (TEM) analysis. Uniform hollow structures were observed with

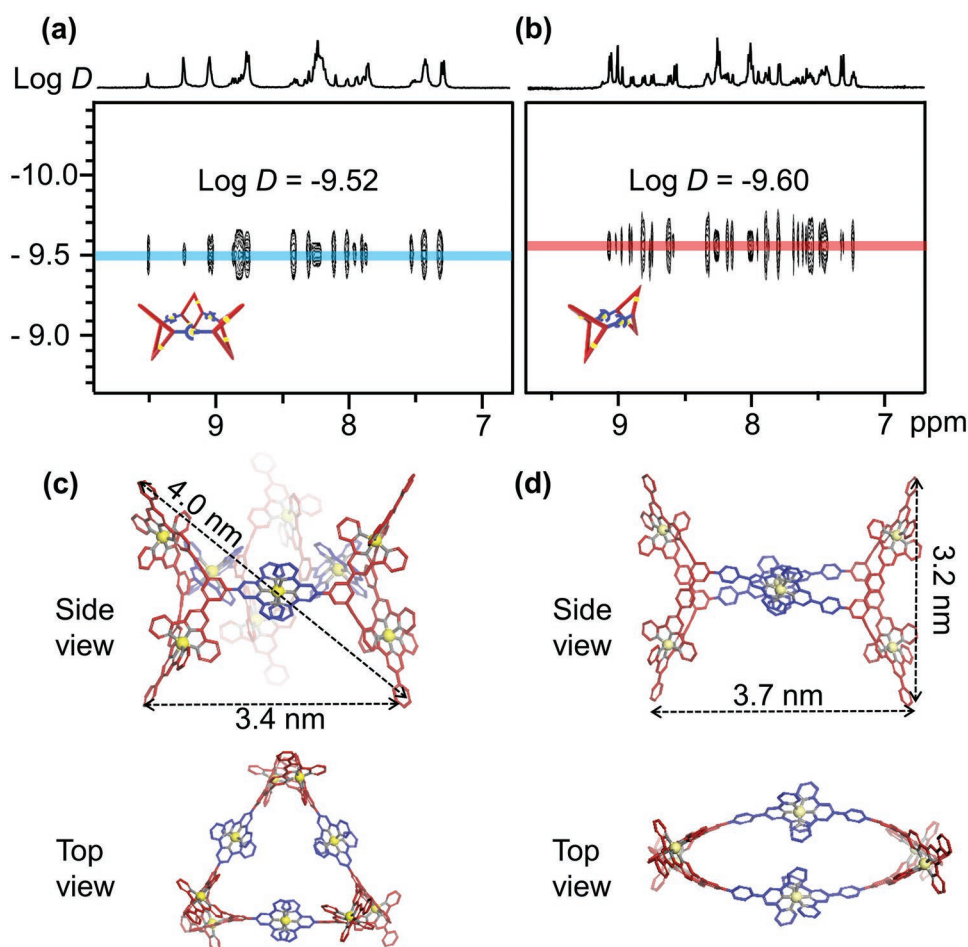


Figure 3. 2D DOSY (500 MHz, 300 K) spectra of a) SA in CD₃CN/CD₃OD (4:1, v/v) and b) SB in CD₃CN. Side view and top view of the representative energy-minimized structures from molecular modeling of c) SA and d) SB, alkyl chains were omitted for clarity.

an outer diameter of 19 nm and an inner diameter ranging from 6–9 nm (Figure 5a and Figure S31, Supporting Information). SB was treated using the same sample preparation protocol as SA. Surprisingly, uniform nanofibers with a diameter of ≈ 7.5 nm were observed (Figure 5b and Figure S32, Supporting Information). Such a great difference inspired us to further examine the impact of different solvent systems on the outcome of hierarchical self-assemblies. When diethyl ether was slowly added into DMF, SA showed nanowire-typed structures (Figure S35, Supporting Information), but SB showed amorphous aggregates (Figure S36, Supporting Information). By switching DMF to acetonitrile, the hierarchical self-assemblies of SA created sheet-like structures (Figure 5c and Figure S33, Supporting Information). However, self-assemblies of SB in this case led to the formation of nanorods with a diameter of ≈ 15.6 nm (Figure 5d and Figure S34, Supporting Information). We attribute the obviously distinct outcomes of hierarchical self-assemblies to the structural differences (e.g., shape, position of alkyl chains) as well as the interactions with solvent systems.^[56]

We further investigated the hierarchical self-assembly behaviors of SA and SB at the liquid/solid interface using STM. We initiated our study with SB. DMF solution

containing SB was drop cast onto a freshly cleaved highly oriented pyrolytic graphite (HOPG) surface for incubation of 15 min. After the incubation, sample was dried with nitrogen gas and used immediately for STM imaging. Very interestingly, nanowire-like pattern extending across a length at micron level was observed (Figure 6a). A zoom-in view of the nanowire revealed a uniform ladder-shaped structure (Figure 6b–f). The average width of the nanowire was measured to be around 3.7 nm, which is consistent with the measured width of SB by molecular modeling (Figure 3d). Based on the superstructure revealed by STM, along with the molecular modeling, we propose that SB might have self-assembled by leaning against each other on the HOPG surface. Such self-assembly behavior might be a result from: 1) π – π interactions between SB and the HOPG surface, 2) high affinity of the HOPG substrate for the alkyl chains,^[57] and 3) intermolecular hydrophobic interactions and CH– π interactions.^[58] Following the same sample preparation protocol, the self-assembly of SA on HOPG was also studied. As can be seen from Figure S37 (Supporting Information), nanowire with hundreds of nanometer length could be observed. The zoom-in images indicate a width of 3.3 nm, which is consistent with the measured size via molecular modeling.

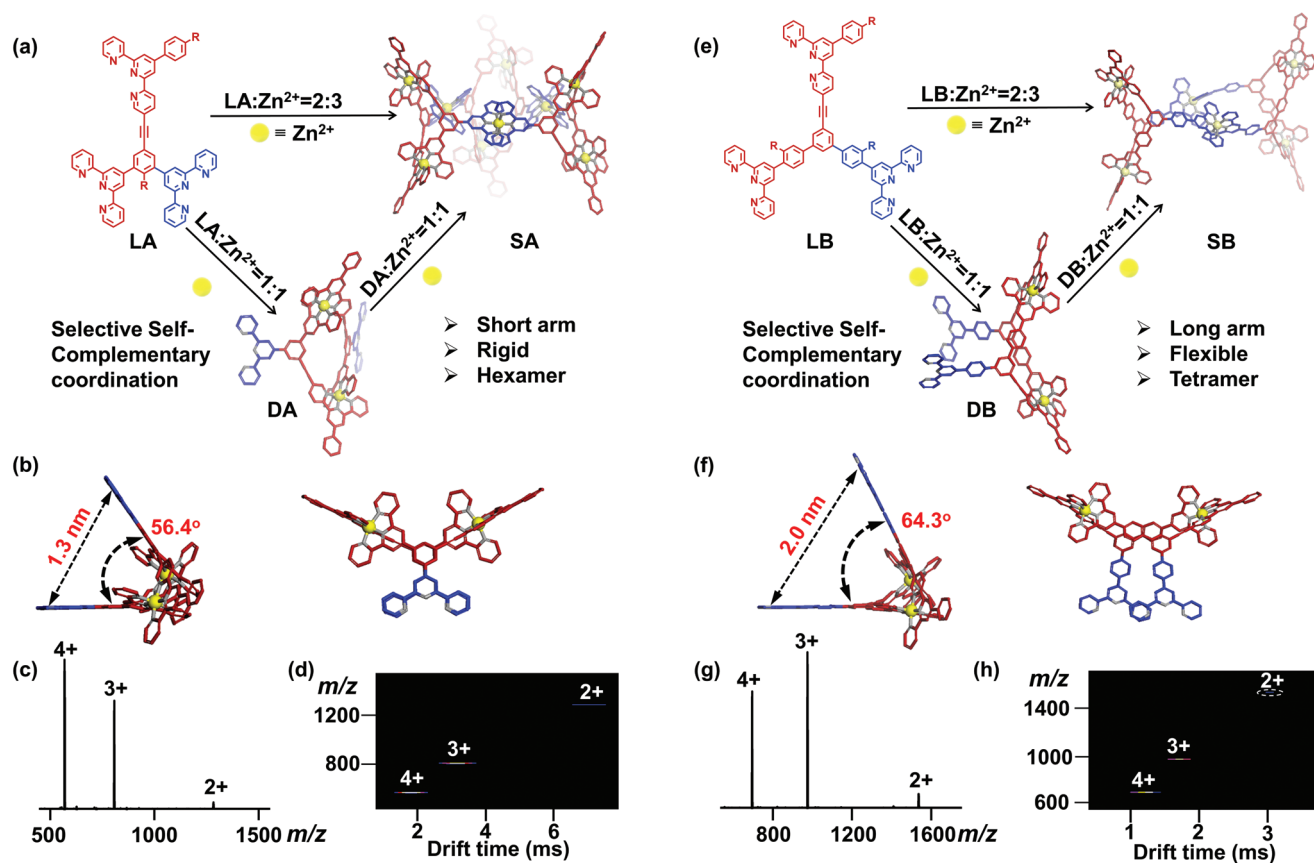


Figure 4. a,e) Self-assembly mechanism of SA and SB. b,f) Different views of geometry-optimized structures of dimers DA and DB. c,g) ESI-MS of dimers DA and DB. d,h) TWIM-MS plots (m/z vs drift time) of DA and DB.

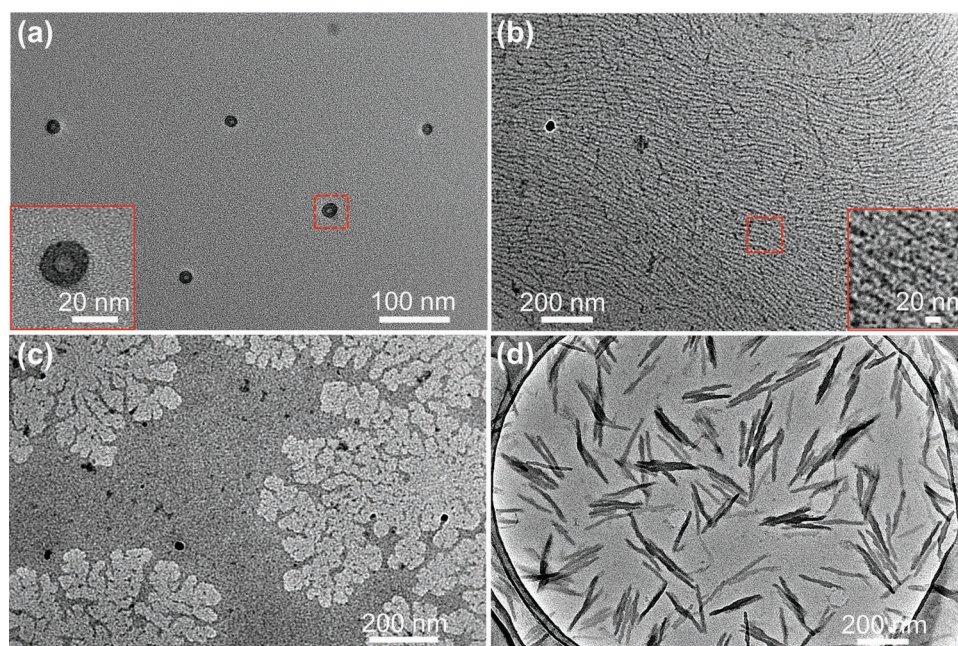


Figure 5. TEM images of a) SA and b) SB, by evaporating THF into DMF. TEM images of c) SA and d) SB, by evaporating diethyl ether into acetonitrile.

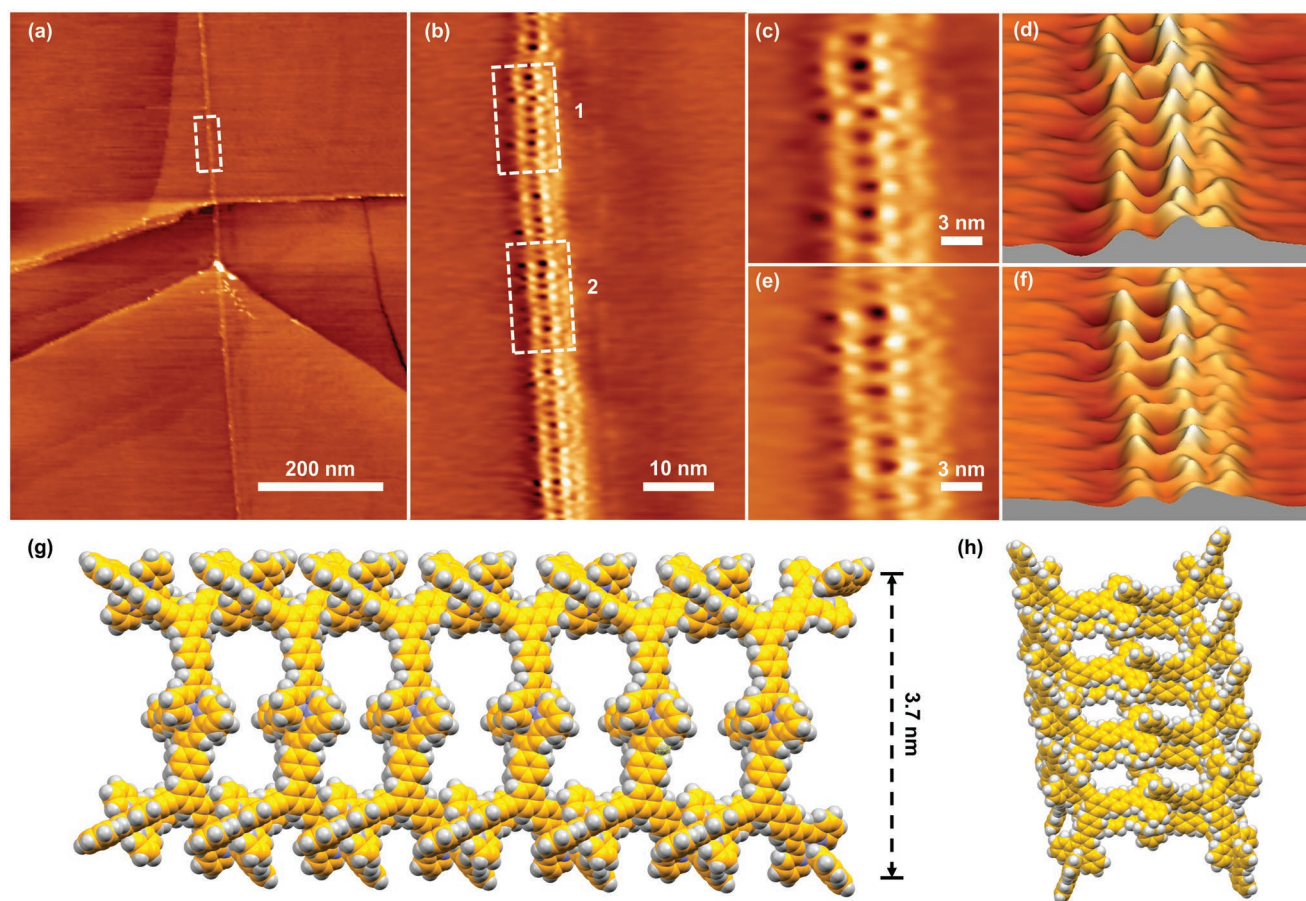


Figure 6. a) Large area STM image of SB self-assembled as nanowire on HOPG surface. b) Zoom-in image of the squared area in (a). c) Zoom-in image of region 1 squared in (b). d) 3D image of (c). e) Zoom-in image of region 2 squared in (b). f) 3D image of (e). g,h) Different views of the proposed packing mode of SB.

We speculate that SA also went through hierarchical self-assembly via the above-mentioned interactions. Such on-surface self-assembly has paved the path for future applications of these metallo-supramolecules in self-assembled and scalable supramolecular devices.

2.5. Investigation of the Photophysical Properties

The photophysical properties of ligands and complexes were investigated by UV–vis absorption and fluorescence spectroscopy. As shown in **Figure 7a**, ligands LA and LB exhibited the prominent absorption peaks at ≈ 280 and 315 nm attributed mainly to π – π^* transition of the tpy groups.^[59] After coordination, the absorption of SA and SB showed a slight red shift, possibly due to the intramolecular charge transfer (ICT).^[60] LA showed strong emission at 383 nm, and the emission of LB appeared at 361 and 377 nm (see **Figure 7b**). The absolute fluorescence quantum yield (Φ_F) values of LA and LB were measured to be 75.33% and 66.50% , respectively. After coordination, the fluorescence of both SA and SB exhibited pronounced red shift, with emission band appeared at ≈ 503 and 487 nm ($\Phi_F = 20.30\%$ and 26.34% ,

respectively), resulting larger Stocks Shifts as compared to the ligands LA and LB.

3. Conclusion

By utilizing the self-complementary motif, we have successfully synthesized two meta-modified terpyridyl ligands LA with short arms and LB with long arms. Hexamer SA and tetramer SB were self-assembled by treating LA and LB with Zn(II) in 2:3 ratio, respectively. The ligands and supramolecules were fully characterized via multi-dimensional NMR spectroscopies and mass spectrometries. By studying the self-assembly processes of SA and SB, we have confirmed that the self-complementary coordinated dimers DA and DB were preferably formed, and the uncoordinated free terpyridines further self-assembled with Zn(II) to form the final structure, in which DA and DB served as edge units. Both SA and SB could go through hierarchical assemblies in solution which produce different morphologies as visualized by TEM. Most importantly, both SA and SB could also form long nanowires via hierarchical self-assembly at the liquid–solid interface. Noticeably, the external conditions play a critical role in

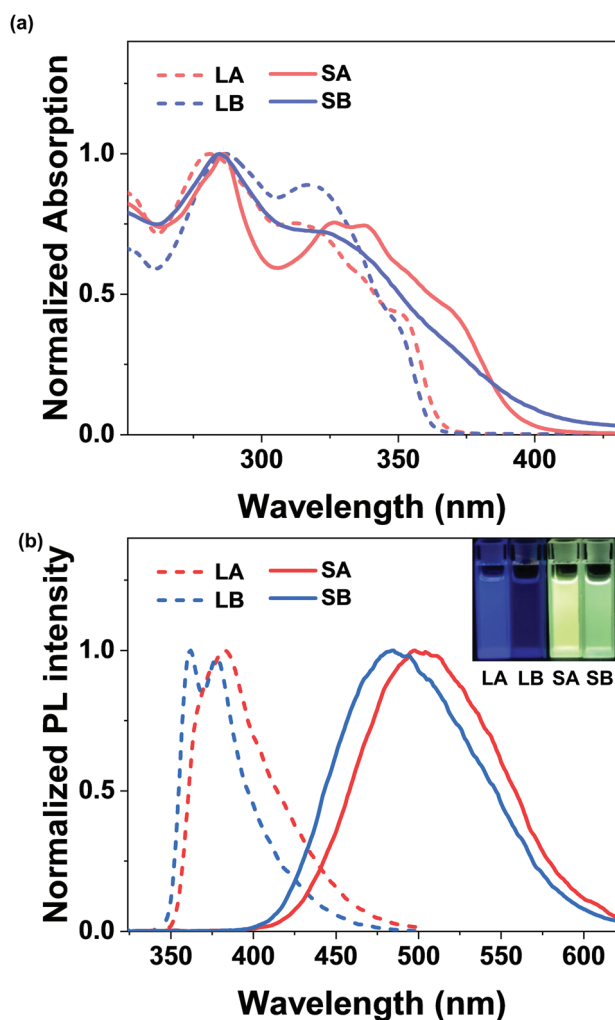


Figure 7. a) Normalized absorption and b) PL intensity ($\lambda_{\text{ex}} = 320$ nm) of ligands in chloroform (10.00 μm) and complexes in acetonitrile (1.00 μm). Inset: photographs of LA, LB, SA, and SB (left to right) under excitation at 365 nm using an ultraviolet lamp at 298 K ($c = 1.00$ μm).

determining the final structure of higher-order assembly, suggesting environmental control as a viable route to tailor the assembly outcome. This work has enriched the family of 3D metallo-supramolecular architectures constructed with terpyridine-based ligands and will further shed light on the applications of metallo-supramolecules in nanoscale, molecular scale devices and fluorescent sensing.

Supporting Information

Supporting Information is available from the Wiley Online Library or from the author.

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Conflict of Interest

The authors declare no conflict of interest.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

Keywords

3D metallo-supramolecules, concave prisms, hierarchical self-assembly, self-coordination, terpyridine

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